

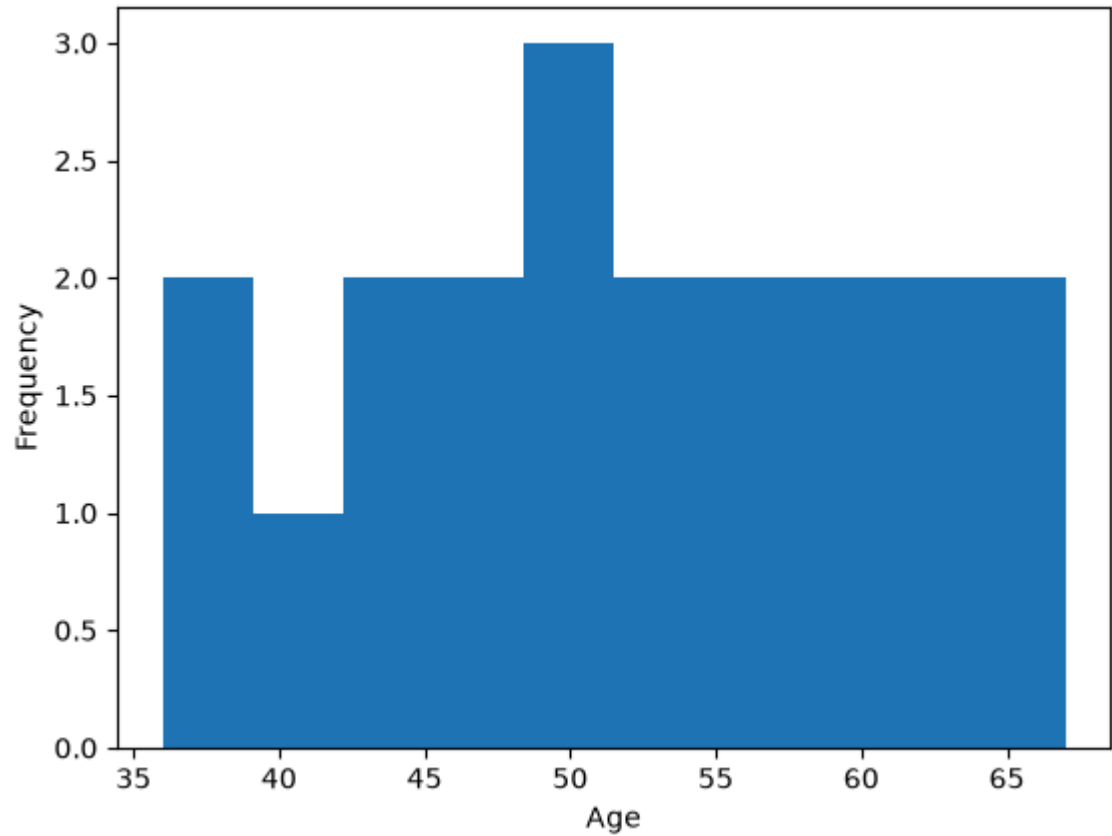
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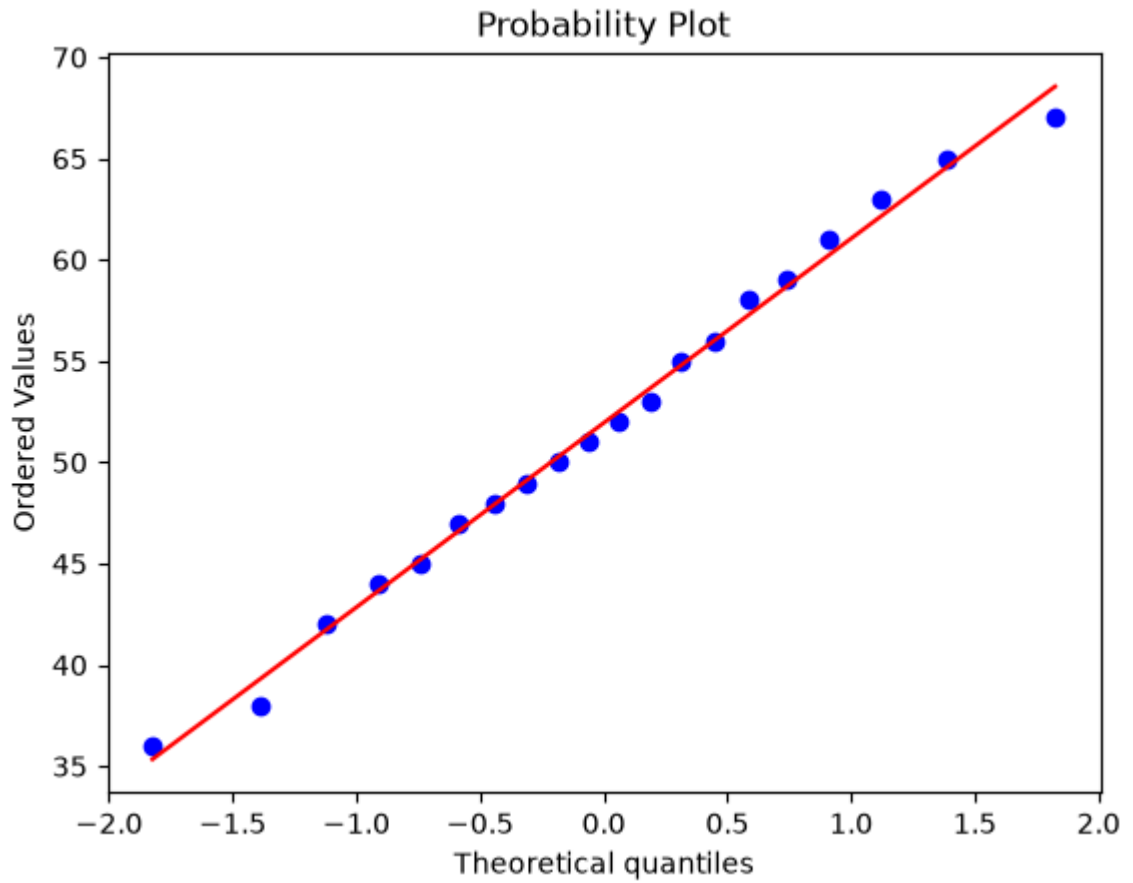
In [15]: import pandas as pd
df=pd.read_csv('breast_cancer_clinical1.csv')
import scipy
from scipy.stats import shapiro
stat,p=shapiro(df['Age'])
print('age:',stat,p)
import matplotlib.pyplot as plt
plt.hist(df['Age'],bins=10)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
import scipy.stats as stats
import matplotlib.pyplot as plt
stats.probplot(df['Age'],dist='norm',plot=plt)
plt.show()
group0=df[df['Lymph_Node']==0]
group1=df[df['Lymph_Node']==1]
import scipy
from scipy.stats import shapiro
stat0,p0=shapiro(group0['Age'])
stat1,p1=shapiro(group1['Age'])
print('Lymph_Node0:',stat0,p0)
print('Lymph_Node1:',stat1,p1)
import matplotlib.pyplot as plt
plt.hist(group0['Age'],bins=10)
plt.xlabel('LN0age')
plt.ylabel('Frequency')
plt.show()
import scipy.stats as stats
import matplotlib.pyplot as plt
stats.probplot(group0['Age'],dist='norm',plot=plt)
plt.show()
plt.hist(group1['Age'],bins=10)
plt.xlabel('LN1age')
plt.ylabel('Frequency')
plt.show()
stats.probplot(group1['Age'],dist='norm',plot=plt)
plt.show()
stat,p=shapiro(df['Tumor_Size'])
print('-----')
print('tumor_size:',stat,p)
group0=df[df['Lymph_Node']==0]
group1=df[df['Lymph_Node']==1]
stat0,p0=shapiro(group0['Tumor_Size'])
print('Lymph_Node0:',stat0,p0)
stat1,p1=shapiro(group1['Tumor_Size'])
print('Lymph_Node1:',stat1,p1)
print('-----')
stat,p=shapiro(df['Ki67'])
print('Ki67:',stat,p)
group0=df[df['Lymph_Node']==0]
group1=df[df['Lymph_Node']==1]
stat0,p0=shapiro(group0['Ki67'])
stat1,p1=shapiro(group1['Ki67'])
print('LN0ki67:',stat0,p0)
print('LN1ki67:',stat1,p1)
print('-----')
df=pd.DataFrame({

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'Variable':['Age','Age','Age','Tumor_Size','Tumor_Size','Tumor_Size','Ki67',  
'Group':['Overall','LN(-)','LN(+)','Overall','LN(-)','LN(+)','Overall','LN(-)',  
'P value':['0.98','0.75','0.88','0.92','0.97','0.85','0.61','0.80','0.99'],  
'Distribution':['Normal','Normal','Normal','Normal','Normal','Normal','Normal','Normal',  
'Next analysis':['t-test','t-test','t-test','t-test','t-test','t-test','t-test','t-te  
})  
print(df)  
df.to_excel('Normality Test Results.xlsx')
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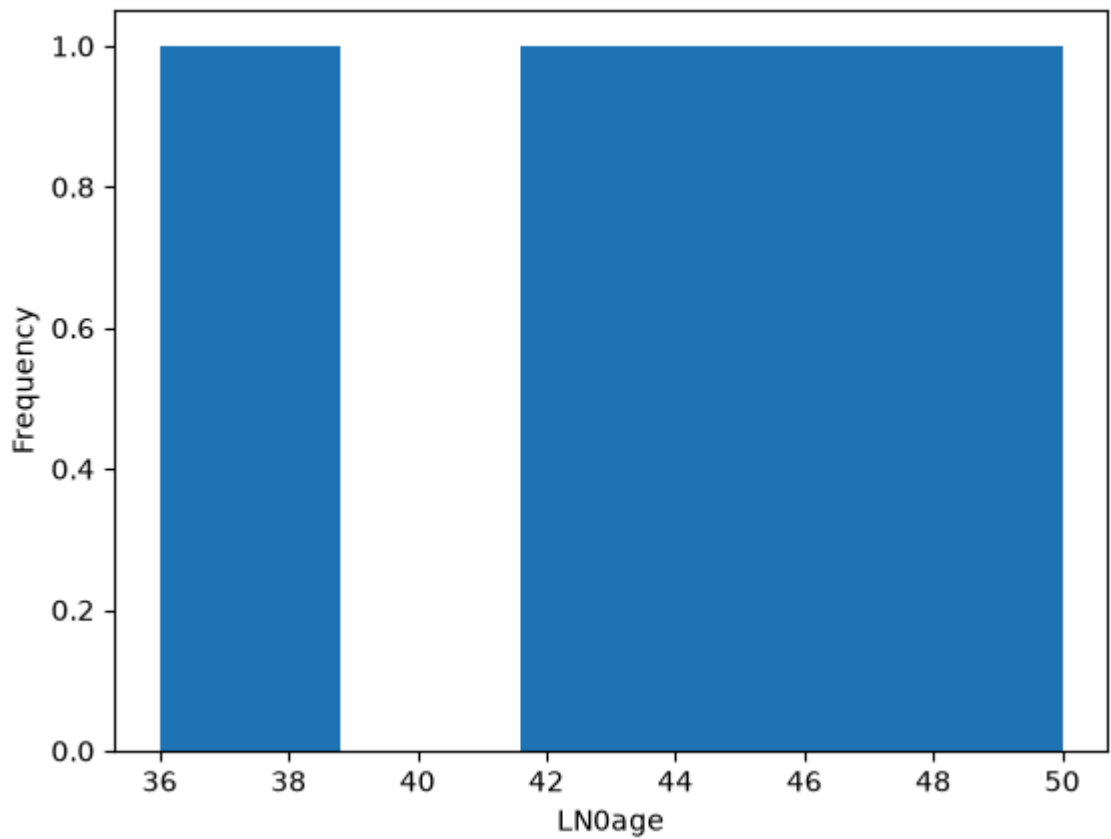
age: 0.9845682420406165 0.9788032525063932

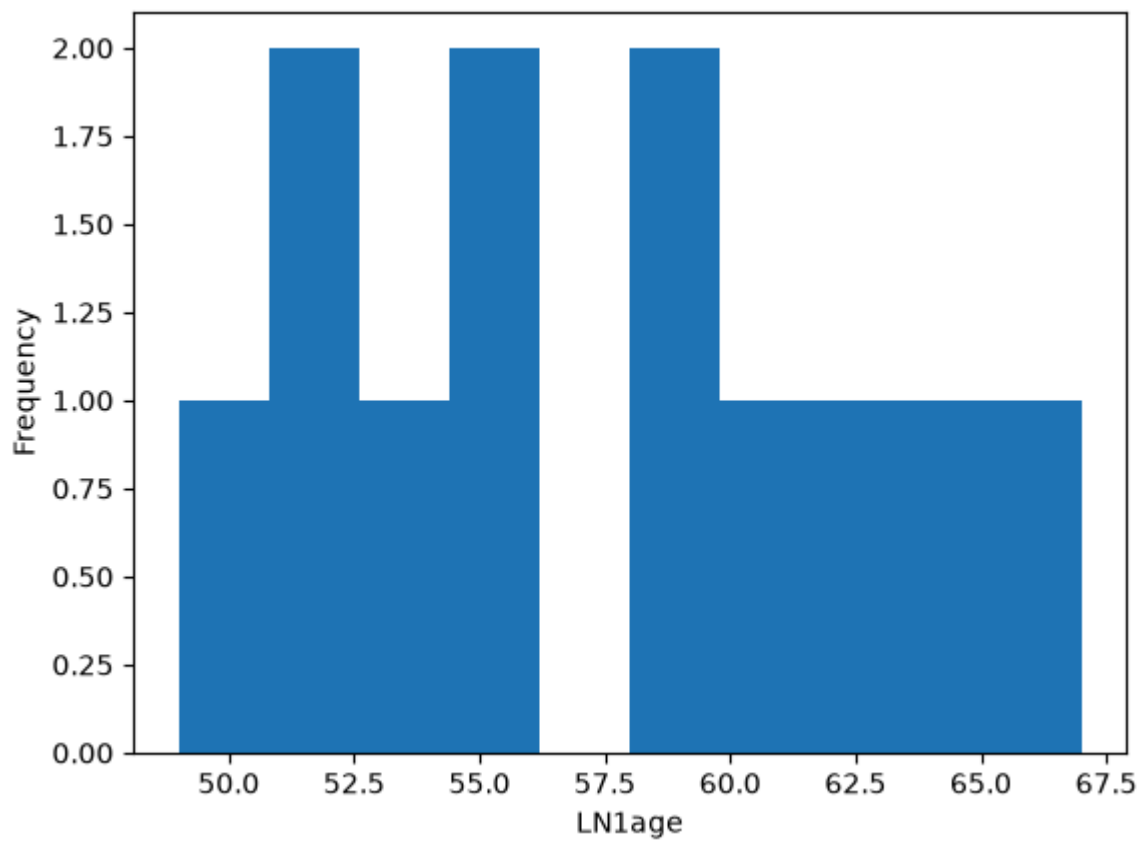
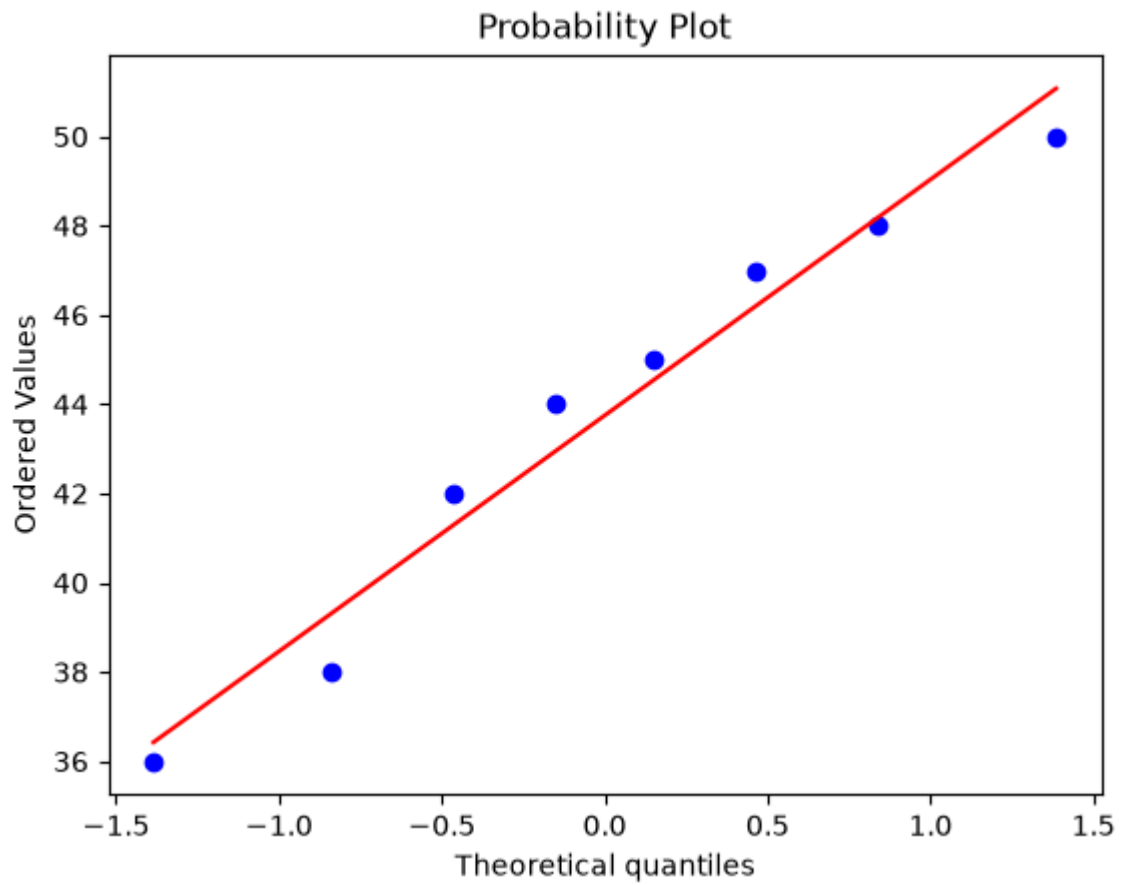


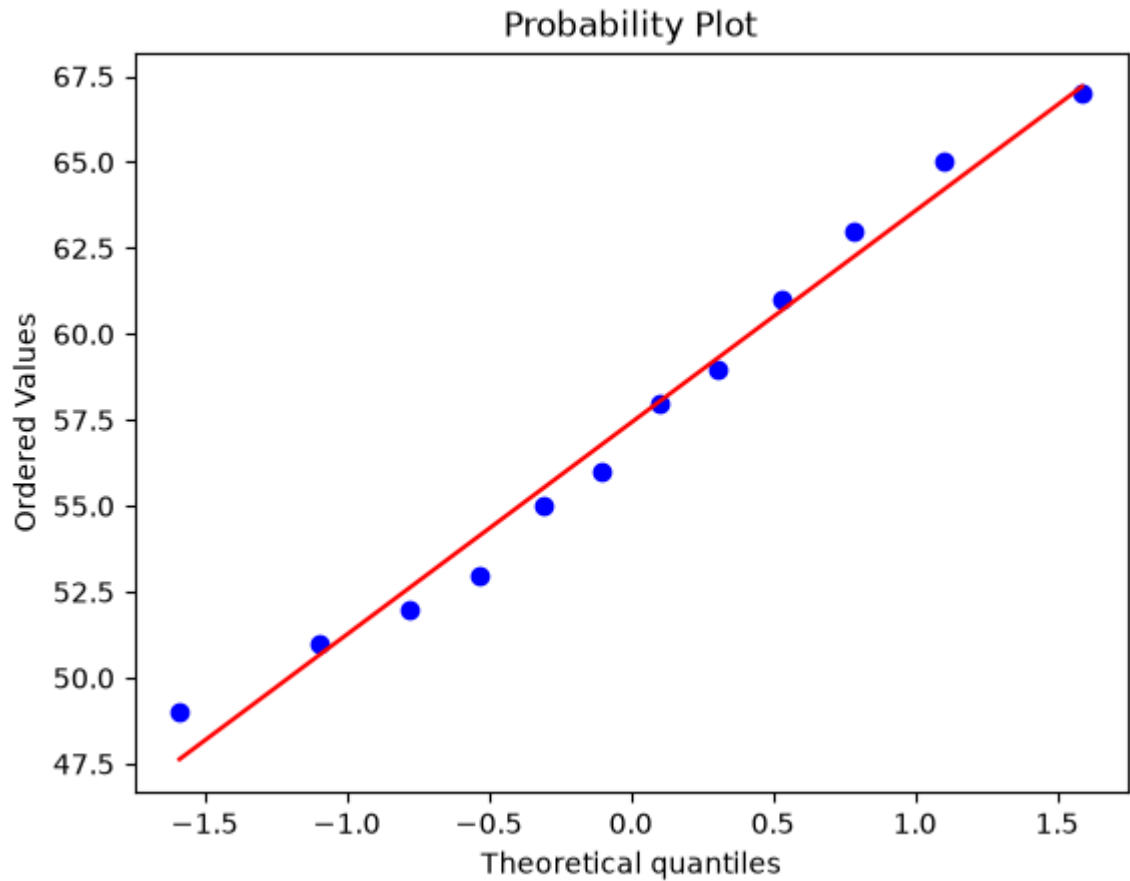


Lymph_Node0: 0.9540916392081262 0.7523161714941502

Lymph_Node1: 0.9676342097425262 0.8845084068009105







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tumor_size: 0.9787044161432614 0.9162922826518878
Lymph_Node0: 0.981909837438283 0.9717312189632211
Lymph_Node1: 0.9649049417779392 0.8508503170903894
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Ki67: 0.9632708807890246 0.6110521265936039
LN0ki67: 0.9590108876742224 0.8006263515466563
LN1ki67 0.98366700211603 0.9943033655535543
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Variable      Group P value Distribution Next analysis
0      Age Overall    0.98      Normal      t-test
1      Age  LN(-)     0.75      Normal      t-test
2      Age  LN(+)     0.88      Normal      t-test
3 Tumor_Size Overall  0.92      Normal      t-test
4 Tumor_Size LN(-)    0.97      Normal      t-test
5 Tumor_Size LN(+)    0.85      Normal      t-test
6      Ki67 Overall   0.61      Normal      t-test
7      Ki67 LN(-)     0.80      Normal      t-test
8      Ki67 LN(+)     0.99      Normal      t-test
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In [ ]:
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In [ ]:
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